ELECTRONIC PRODUCT DESIGN & REVERSE ENGINEERING

PROJECT PART B

Simulation and Study of PN Junction and Zener Diode Characteristics

TINKERCAD CIRCUIT SIMULATION REPORT

GROUP 51

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1. Aim and Objectives

To simulate and study forward and reverse current-voltage characteristics of a PN junction diode and a Zener diode using Tinkercad Circuits, compare their behaviour, and justify the use of PN diodes as switches and Zener diodes as voltage regulators.

Objectives

- Build forward- and reverse-biased diode circuits.
- Record diode voltage and current at multiple source settings.
- Plot I–V characteristics and identify the forward knee and Zener breakdown regions.
- Build a Zener shunt regulator and examine output-voltage stability.

2. Components Required

Component	Suggested value/type	Purpose
DC power supply	Adjustable 0–10 V or suitable range	Vary input between readings
PN diode	1N4007 or available silicon diode	Forward/reverse characteristics
Zener diode	5.1 V nominal, if available	Breakdown and regulation
Series resistor	1 kΩ	Limits current
Load resistor	2 kΩ	Regulator load
Multimeters	Two if available	Measure voltage/current
Breadboard and wires	As needed	Circuit construction

3. Safety and Measurement Notes

Use a series resistor in every diode test. Connect the ammeter in series and the voltmeter in parallel across the diode or load. Stop the simulation before rewiring or changing source settings. An ordinary PN diode model is not a substitute for a Zener model; it generally will not reproduce designed reverse breakdown. Do not force an ordinary diode into breakdown.

Observation note: The numerical readings presented in the observation tables are the experimental observations recorded from the Tinkercad circuit simulations. The separate sample calculation (Section 9) is provided for illustration and is not a replacement for the recorded observations.

4. Circuit A — PN Diode Forward Bias

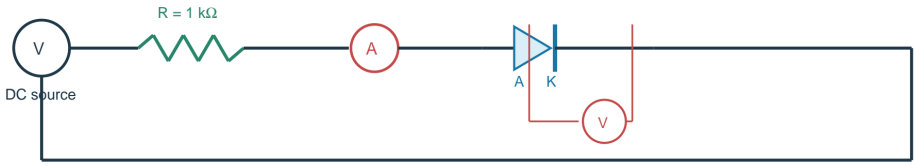


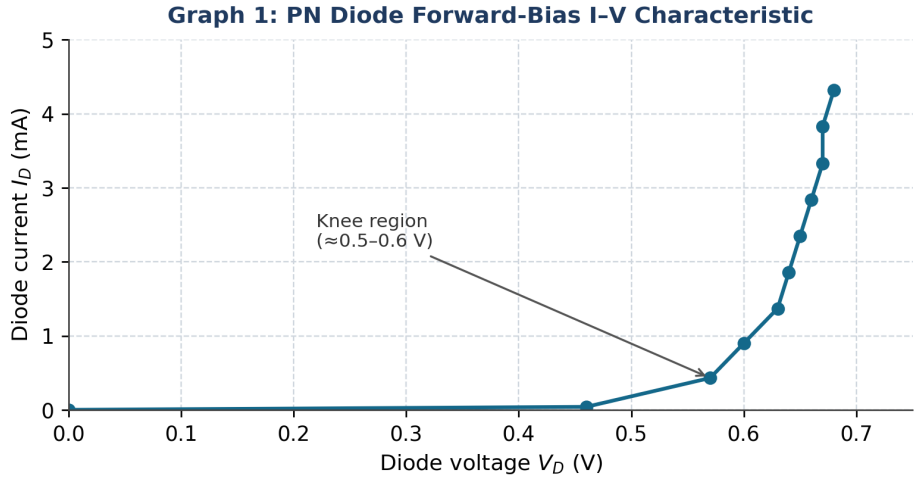
Figure — Circuit A: PN diode forward bias. Ammeter connected in series; voltmeter connected in parallel across the diode.

Connections: Power supply positive → 1 kΩ resistor → ammeter → diode anode; diode cathode → supply negative. Connect the voltmeter across the diode.

Procedure: Start at 0 V, run the simulation and record diode voltage/current. Stop, set the next source voltage, and repeat. Use 0, 0.5, 1.0 … 5.0 V if supported. Record the actual diode voltage, not just the supply setting.

Observation Table A — Forward-bias readings

Supply V_S (V)	Diode voltage V_D (V)	Diode current I_D (mA)
0.0	0.00	0.00
0.5	0.46	0.04
1.0	0.57	0.43
1.5	0.60	0.90
2.0	0.63	1.37
2.5	0.64	1.86
3.0	0.65	2.35
3.5	0.66	2.84
4.0	0.67	3.33
4.5	0.67	3.83
5.0	0.68	4.32



Graph 1 — PN diode forward-bias I–V characteristic. The recorded current is only 0.04 mA at 0.46 V, but rises to 0.43 mA at 0.57 V and then steeply to 4.32 mA at 0.68 V. The diode voltage changes by only about 0.11 V while the current increases roughly ten-fold, showing the forward knee at about 0.5–0.6 V.

5. Circuit B — PN Diode Reverse Bias

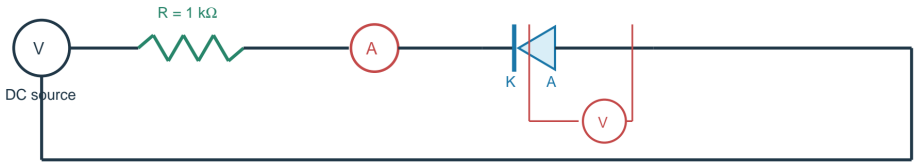


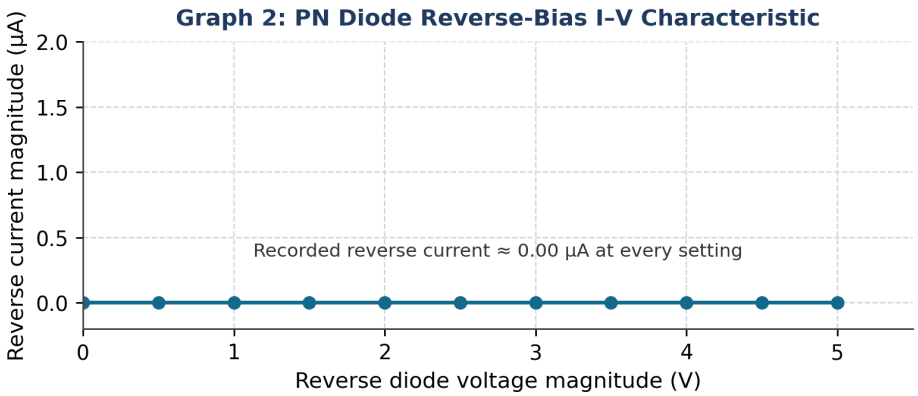
Figure — Circuit B: PN diode reverse bias. Ammeter connected in series; voltmeter connected in parallel across the diode.

Connections: Keep the 1 kΩ resistor and measurement arrangement. Reverse the diode so its cathode faces the positive supply and its anode faces the negative supply. The voltmeter remains across the diode.

Procedure: Repeat the source settings used for Circuit A. Record reverse-voltage and reverse-current magnitudes, or use signed values consistently. Reverse current should be very small at low voltages.

Observation Table B — Reverse-bias readings

Supply magnitude (V)	Reverse diode voltage magnitude (V)	Reverse current (µA)
0.0	0.00	0.00
0.5	0.50	~0.00
1.0	1.00	~0.00
1.5	1.50	~0.00
2.0	2.00	~0.00
2.5	2.50	~0.00
3.0	3.00	~0.00
3.5	3.50	~0.00
4.0	4.00	~0.00
4.5	4.50	~0.00
5.0	5.00	~0.00



Graph 2 — PN diode reverse-bias I-V characteristic. In contrast to the forward-bias case (Circuit A), the recorded reverse current magnitude remains approximately zero (~0.00 µA) from 0 V to 5 V, so the plotted trend is a flat line along the voltage axis. The diode blocks essentially all current in reverse bias over the tested range, and breakdown was not forced in this ordinary diode. A real diode has a very small leakage current, which is below the meter's resolution here.

6. Circuit C1 — Zener Forward Bias

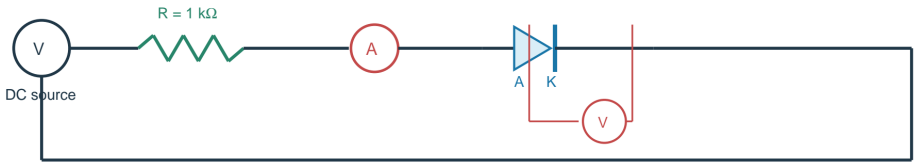


Figure — Circuit C1: Zener diode forward bias. Ammeter connected in series; voltmeter connected in parallel across the Zener diode.

Connections: Replace the PN diode with a Zener model, keeping the 1 kΩ series resistor. In forward bias, connect the anode to the positive side and the cathode to the negative side. The ammeter is in series and the voltmeter is across the Zener diode.

Observation Table C1 · Zener forward-bias readings

Supply V_S (V)	Zener forward voltage V_F (V)	Forward current I_F (mA)
0.0	0.00	0.00
0.5	0.46	0.04
1.0	0.57	0.43
1.5	0.60	0.90
2.0	0.63	1.37
2.5	0.64	1.86
3.0	0.65	2.35
3.5	0.66	2.84
4.0	0.67	3.33
4.5	0.67	3.83
5.0	0.68	4.32

7. Circuit C2 — Zener Reverse Breakdown

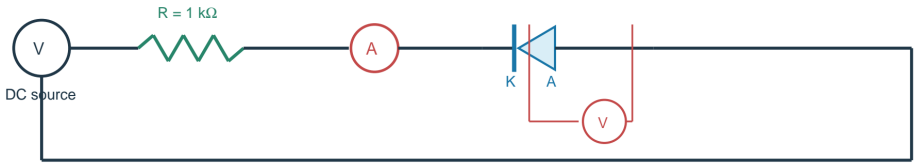
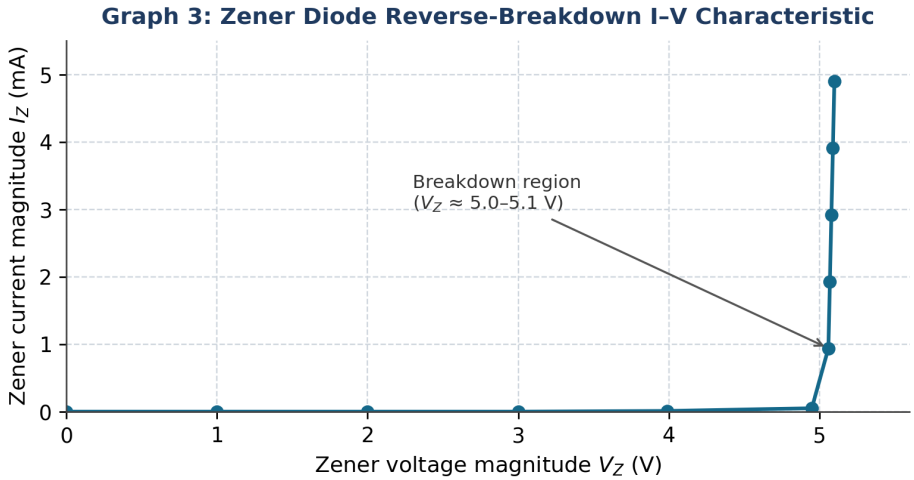


Figure — Circuit C2: Zener reverse-breakdown test. Ammeter connected in series; voltmeter connected in parallel across the Zener diode.

Connections and procedure: Reverse the Zener so its cathode faces the positive supply. Use a model that supports reverse breakdown. Increase the source carefully toward the nominal Zener voltage, recording extra readings near the knee. For a 5.1 V Zener, a 0–10 V supply and 1 kΩ series resistor are a starting setup, subject to the model and ratings.

Observation Table C2 — Zener reverse-breakdown readings

Supply V_S (V)	Zener voltage V_Z (V)	Zener current I_Z (mA)
0	0.00	0.00
1	1.00	0.00
2	2.00	0.00
3	3.00	0.00
4	3.99	0.01
5	4.95	0.05
6	5.06	0.94
7	5.07	1.93
8	5.08	2.92
9	5.09	3.91
10	5.10	4.90



Graph 3 — Zener diode reverse-breakdown I-V characteristic. The Zener current is essentially zero (0.05 mA or less) up to 4.95 V. Between 5.06 V and 5.10 V the current rises sharply from 0.94 mA to 4.90 mA, about 0.99 mA for every 0.01 V, while the voltage stays almost constant near the nominal 5.1 V. This sharp knee is the reverse-breakdown region of the Zener diode.

8. Circuit D — Zener Shunt Voltage Regulator



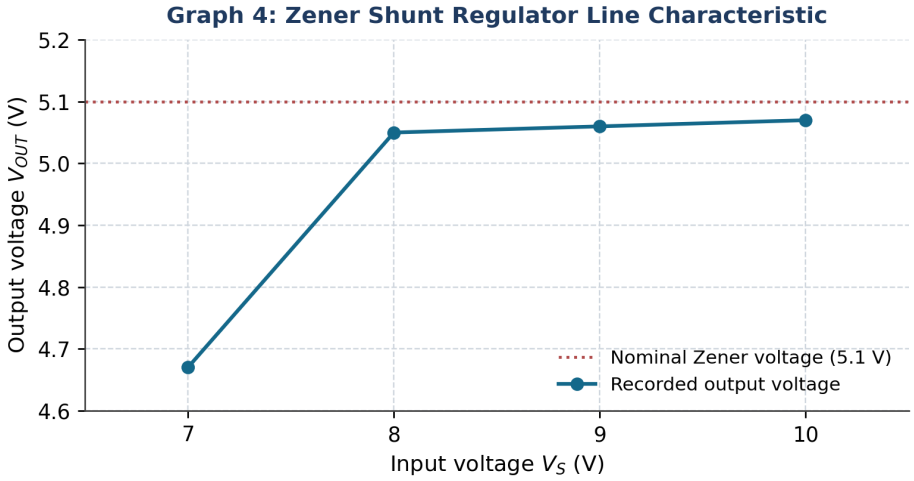
Figure — Circuit D. Zener cathode connected to the output-positive node; anode and load return connected to ground.

Connections: The input supply feeds the output node through R_S = 1 kΩ. Connect the Zener cathode to the output node and its anode to ground. Connect R_L = 2 kΩ from the output node to ground. Measure output voltage across the load/Zener.

Procedure: Increase the input in steps such as 7, 8, 9 and 10 V. Record output voltage. Measure or calculate load and Zener currents if possible. Regulation occurs only while Zener current remains within its valid operating range.

Observation Table D — Regulator readings

Input V_S (V)	Output V_{OUT} (V)	Load current I_L (mA)	Zener current I_Z (mA)
7	4.67	2.33	0.00
8	5.05	2.53	0.42
9	5.06	2.53	1.40
10	5.07	2.54	2.39



Graph 4 — Zener regulator line characteristic. At V_S = 7 V the output is 4.67 V and the recorded Zener current is 0.00 mA, so the Zener is not yet conducting. From 8 V to 10 V the output stays between 5.05 V and 5.07 V, a change of only 0.02 V for a 2 V change in input, slightly below the nominal 5.1 V. This shows regulation once the Zener is in breakdown.

9. Sample Calculation — Illustration Only

> **Note:** This section is an idealized illustrative calculation. It is separate from the recorded observation tables and does not replace any recorded Tinkercad reading.

Assume an idealized 5.1 V Zener regulator with $V_S = 9$ V, $V_Z = 5.1$ V, $R_S = 1$ kΩ and $R_L = 2$ kΩ.

Series current: $I_S = (V_S - V_Z) / R_S = (9 - 5.1) / 1000 = 3.9$ mA.

Load current: $I_L = V_Z / R_L = 5.1 / 2000 = 2.55$ mA.

Zener current: $I_Z = I_S - I_L = 3.9 - 2.55 = 1.35$ mA.

This assumes the Zener remains in regulation. Check the model's knee current before treating this as an operating point.

10. PN Diode vs Zener Diode

Feature	PN junction diode	Zener diode
Forward bias	Conducts after the knee region	Similar forward conduction
Reverse bias before breakdown	Very small leakage current	Very small leakage current
Breakdown	Normally avoided in ordinary operation	Designed for controlled reverse breakdown
Typical use	Rectification, switching, clipping	Voltage reference and shunt regulation
Voltage behaviour	Not normally used for reverse voltage regulation	Approximately constant near V_Z within its operating range

11. Justification of Applications

PN diode as a switch: In forward bias, the diode conducts and behaves approximately like an ON switch after sufficient forward voltage is applied. In reverse bias, it blocks most current and behaves approximately like an OFF switch. Real diodes have a forward drop and nonzero leakage.

Zener diode as a voltage regulator: In reverse breakdown, the Zener holds its voltage approximately constant across a load. A series resistor limits current and absorbs changes in input voltage. Regulation is maintained only while Zener current and power remain within rated limits.

12. Precautions

1. Check anode/cathode orientation before every run.
2. Include a current-limiting resistor.
3. Put the ammeter in series and voltmeter in parallel.
4. Stop simulation before rewiring or adjusting the source.
5. Use a Zener model that supports breakdown.
6. Save screenshots of each working circuit and readings.
7. Complete tables and graphs using actual observations.

13. Conclusion

The recorded observations show that the PN diode conducts readily in forward bias, with the current rising from 0.04 mA at 0.46 V to 4.32 mA at 0.68 V, and blocks current in reverse bias, with the reverse current recorded as approximately zero from 0 V to 5 V. The Zener diode, tested in reverse, carried almost no current up to 4.95 V and then entered breakdown near 5.06–5.10 V, where the current rose from 0.94 mA to 4.90 mA while the voltage stayed almost constant. In the shunt regulator (Circuit D), the output was 4.67 V at a 7 V input and stayed between 5.05 V and 5.07 V for inputs of 8–10 V, close to the nominal 5.1 V. These results explain PN diode switching and Zener voltage regulation. In forward bias, the Zener diode exhibited characteristics similar to the PN junction diode under the tested simulation conditions.